

# Special Interest Group: The Present and Future of Esports in HCI

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## ABSTRACT

Rapidly gaining in mainstream appeal, esports constitute a phenomenon at the intersection of different Human-Computer Interaction (HCI) perspectives—relating to games user research, multifaceted aspects of game design for performance and entertainment, design and support of social or interpersonal interaction, inclusion vs. toxicity in online communities, visualization of esports data, subdomains like physical esports, as well as connections with education and health contexts. This special interest group (SIG) will provide a space for HCI researchers and practitioners to connect and discuss themes at the intersection of HCI and esports. It will serve as a starting point for mapping the esports design and research landscape in order to identify and pursue opportunities for research, to increase awareness for collaboration in this domain within the HCI community, to share experiences and knowledge, and to establish a community to shape the future of esports.

## CCS CONCEPTS

• **Human-centered computing** → **Human computer interaction (HCI)**; • **Applied computing** → **Computer games**.

## KEYWORDS

esports, communities, spectatorship experience, health, training, education, mixed reality, streaming

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## 1 BACKGROUND

Over the last years, esports – competitive gaming (e.g., Strategy, MOBA, Simulation) – has become a booming business attracting million of viewers and players. As increasing numbers of researchers within the CHI community are involved in esports or work related to esports, this SIG will provide a starting point for a timely discussion. The goal of the SIG is to identify intersections between HCI and esports and to establish holistic discussions about

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relevant themes in this area to strengthen our community's research activities in a way that benefits both the field and research within it.

Based on initial discussions between the researchers aiming to establish this SIG, the following themes were identified as starting points for future research:

*Streamer and Spectatorship Experience.* The rise of online streaming and esports events has turned spectating other people playing video games into a mass phenomenon [13]. The spectatorship can take on different forms (e.g., online versus on-site) which can be influenced by various factors (cf. [23]). The CHI2020 workshop *Be Part Of It: Spectator Experience in Gaming and Esports* [18] was a first step in this direction aimed at fostering discussions on how spectators can take a more active role in order to improve the spectator experience. It became clear that new design approaches that consider the different motivation and needs of streamers and spectators are necessary. Given the different crisis we face today – e.g., climate or disease – a question for future will also be on how the recreate the on-site atmosphere in a spectator's home and how the communication between players and spectators and between spectators can look like.

*Esports Communities.* The social influence of gameplay has been studied extensively in games user research; consequently, it is broadly understood that online gaming can facilitate and support meaningful social networks and interpersonal relationships [26]. Parallel to this – and less widely studied – are the communities and interpersonal relationships that emerge in esports contexts. Extant research supports the importance of relationships that develop in professional play contexts [12], and establishes social motivations as a primary incentive for viewer participation in livestreams [13]. There are notable opportunities for exploring the social networks that arise in specific esports contexts (e.g., fandoms that emerge around teams or personalities), as well as the possibilities for leveraging these networks to support wellbeing and prosocial behaviours. These inquiries are also critical in better understanding antisocial behaviours (i.e., toxicity) endemic to competitive gaming contexts, which – while often normalised by players – represent a barrier to inclusion and a threat to player wellbeing in esports [27]. There is a need for further studies that identify and test design principles to foster social connection and inclusivity in esports.

*Inclusion in Esports.* Despite its stratospheric rise and broad appeal, diversity and representation in esports is still notably limited. Research in esports and online gaming contexts has identified toxicity as a central contributing factor, finding that women and minorities such as Hispanic and African American players experience significantly more targeted harassment than other groups [2, 19]. To this end, targeted players describe withdrawing from the space altogether (or indicate a reticence to join to begin with) [10]. Another significant barrier that prevents equality in esports (from entry to retention) relates to material means (e.g., access to particular games, gaming equipment, and reliable internet). In summary, there is a vital and distinct opportunity in CHI to explore, address, and inform improvements to inclusivity and representation in esports. In particular, the generation and dissemination of concrete solutions and

guidelines for promoting inclusion represent an important avenue that would benefit from scholarly investigations.

*Physical Esports.* Today, technologies and approaches such as immersive Virtual and Augmented Reality allow players to physically engage in games [14, 20, 22]. Esports is following this trend, and thus more and more physical game genres (e.g. VR-games and exergames) are establishing themselves in professional leagues [21]. However, little is known about the required features of such technologies when used in competitive gaming, as well as about specific physical, cognitive and mental requirements of athletes performing this hybrid combination of digital and analogue sports. Therefore, there is a huge potential in HCI research and design to reveal untapped potentials in this promising interdisciplinary field.

*Physical, Cognitive and Mental Health in Esports.* Esports athletes are regularly exposed to a wide variety of physical, cognitive and mental stresses during daily training and competitions. The average esports athlete spends numerous hours a day sitting in front of a screen, playing and practicing their games. The lack of physical activity, the high cognitive and mental load, and the mostly one-sided physical posture involved often lead to physical, cognitive and mental health problems. A current study recommends developing and incorporating a health management model for esports athletes as well as defining and understanding the medical needs of esports athletes as it is standard for other athletes [16]. HCI research and design could help exploring specific prevention and rehabilitation tools and methods to help maintain and improve health of esports athletes.

*Performance Enhancement in Esports.* The abovementioned health impairments of esports athletes often lead to reduced performance and a high drop-out rate especially among professional esports athletes. To counteract these health problems and to achieve the peak performance necessary for competition, esports athletes need holistic training (and recovery) that is tailored to their needs and optimally complements their game genre-specific training. Current findings from sports science emphasize, among other things, that strengthening physical, cognitive, and mental performance resources is essential and forms the basis for healthy and high-performing esports athletes [21]. At the same time, there is still an existing gap in the right training methods, since the esports-specific needs and requirements differ from those of traditional athletes. Hence, new training methods (e.g., combined physical-cognitive training [21]) and tools (e.g., attractive and effective sports tech- or game-based approaches [21]) need to be explored. Important existing research and design questions relate to how multimedia platforms can be designed to optimize performance enhancement (both individual and group), in a holistic way that encompasses all its relevant aspects.

*Esports Visualizations.* Visualization can help to review and interpret in-game performances, e.g., to understand how players played the game or to identify new game strategies (cf. [1, 28]). As games can be complex, one challenge is to offer and develop interactive visualizations depending on the motivations and needs of the different stakeholders. For example, to support players' and coaches' with analysis and training tasks or to help spectators compare different matches in a meaningful manner (e.g. by using dashboards [5]).

Another challenge is how visualizations can be used in real time as in-game visualizations. Such visualizations can be helpful for less experienced players for their training or to provide opportunity that they can play with more experienced players at the same level. However, there is also a need to define under which conditions assistance is permitted as well as how and what information should be visualized in order to avoid cheating.

*Modeling and Strategy Identification of Esports Behaviors.* One other aspect of esports is the use of large datasets to model team behaviors as well as spatio-temporal decision-making. This type of data provide granular details on users' behaviors as well as contextual details within the virtual environment that drove these behaviors. There has been several works already leveraging such data to investigate spatio-temporal behaviors (e.g., [9]), team behaviors (e.g., [8]), and modeling of problem solving behaviors (e.g., [1, 3]). Such work is at the intersection of HCI and AI and brings up a very important and interesting investigation that crosses between these two fields. This intersection is important and highly relevant for CHI community and can be leveraged with visualization for a closer and deeper understanding of the User Experience as well as questions around explainable AI.

*Esports and Education.* Esports became a viable educational tool in both K-12 and higher education, preparing students for future careers [7]. Some schools utilize esports to engage and recruit students and establish esports clubs which provide students with a variety of learning and engagement opportunities in a peer-to-peer environment. As outlined above, esports games provide players with tremendous amount of data to assist their strategy development. This has implications for improving players' data literacy skills as prior studies in games research showed evidence that players use statistics provided by game companies to improve their strategies. Players also apply theories to their game strategies which can strengthen their scientific reasoning, metacognitive skills and collaborative problem solving capabilities. As a form of videogaming, esports acts like a platform for students to develop better communication skills both online and offline in a competitive gaming context [11] and improve their cognitive skills, especially spatial cognition [4]. Further studies are needed to explore the role of esports in education.

*Team Formation and Interpersonal Dynamics in Esports.* Team performance, formation and interpersonal dynamics are a central aspect in esports research [6, 8]. It concerns, among others, a unique interplay in interpersonal action, coordination, assessment and decision making, in high pressure environments [11]. Important research/design questions for on/offline team formation and coordination prevail, such as e.g. how to best support strategy development through multimodal communication, how to build towards optimal performance and team synergy, and reinforcing personal relationships [11]. These inquiries are critical not only to organizational success, but to supporting player wellbeing and mental health.

*Multimodal Factors: Sound Effects, Music, and Haptics.* Multimodal aspects relating to audio and haptics in esports are largely unexplored. Sound effects provide a response to player actions, as an important method of conveying feedback – in competitive

games, many players turn off music to better concentrate on sound effects [25]. This function of sound effects is also strongly prevalent in physical exergames [20]. In contrast, music has an ambivalent status: In PC-based games, it can increase immersion [24], but in games that feature more embodied interaction (e.g., in virtual reality, or augmented-reality exergames), music is often not perceived consciously, and does not seem to impact immersion [20] (despite its prominent ability to improve mood and performance, as well as reduce perceived exertion during physical exercise [15]). Future studies and interdisciplinary perspectives are needed to explore functions of audio in competitive games and esports. Finally, the haptic experience of a game (in conventional or physical esports) could similarly play strong role in conveying feedback to players [17].

## 2 AGENDA

The SIG meeting will start with an introduction and an overview about the identified themes. The presentation will be followed by a discussion to invite new perspectives and identify further topics. We plan break-out rooms to allow for more in-depth focus on specific themes depending on participants' interests. Afterwards, each group will report back their insights and results to the whole group for cross-group discussion.

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